MILITARY STANDARD

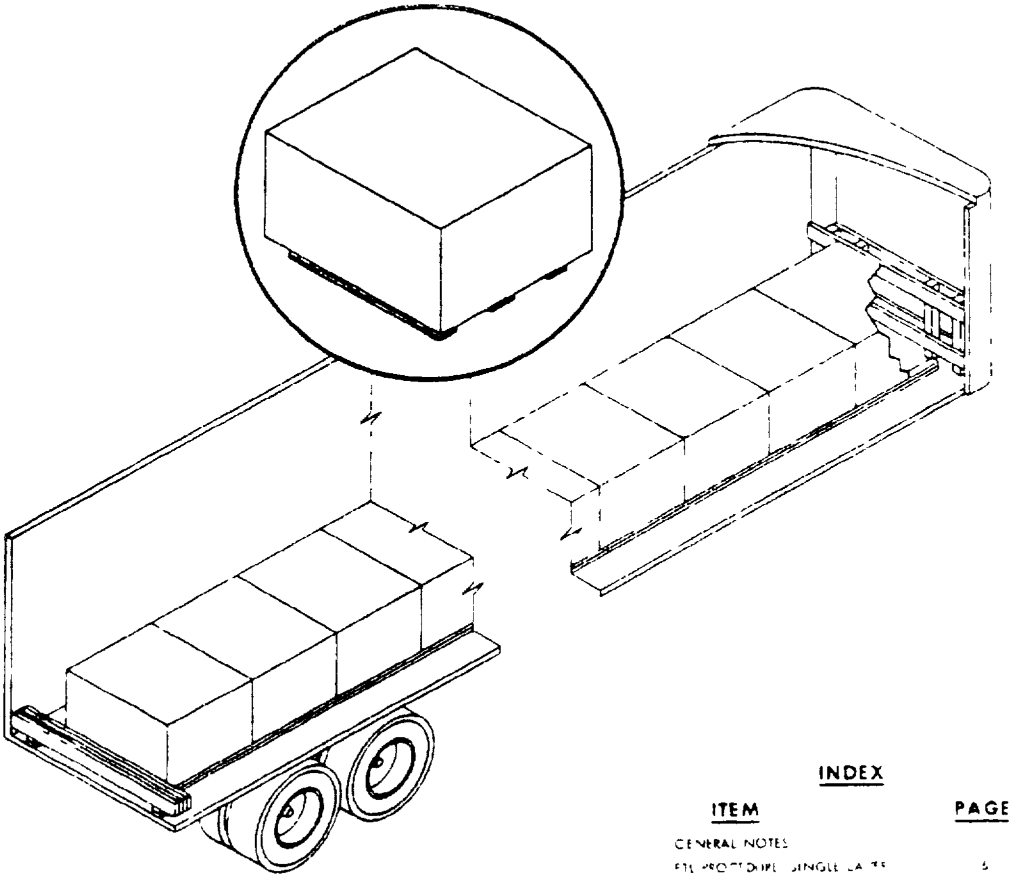
TRUCKLOADING OF HAZARDOUS MATERIALS

PALLETIZED UNIT LOADS
SINGLE ROW PATTERN

MIL-STD-1320-1
(NAVY)

29 JANUARY 1975

[illegible]
[illegible]
22 AUGUST 1967



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NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINER LOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK 136

FSC 8140

AUTHORIZED AND RELEASED
FOR HIGHWAY SHIPMENTS
ONLY

-29
SIGNATURE, TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE [illegible] DATE

ORIGINATOR ______________________
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE NEW JERSEY

PAGE 1 OF 9

GENERAL NOTES

THE INDEX TO STANDARDS FOR PALLETIZING TRUCKLOADING RAILCAR LOADING AND CONTAINER LOADING MIL HDBK 236 (NAVY) LISTS HAZARDOUS MATERIALS AND INDICATES THE CORRECT DOCUMENT TO BE USED IN TRUCKLOADING OF EACH ITEM INDEXED MIL-HDBK-236 INDEXES THE UNIT LOADS THAT ARE AUTHORIZED TO BE TRUCKLOADED IN ACCORDANCE WITH THIS DOCUMENT

2 THIS DOCUMENT PROVIDES DETAILED TRUCKLOADING INSTRUCTIONS APPLICABLE TO PALLETIZED UNIT LOADS WHEN THE SUM OF THE LENGTH AND WIDTH OF A UNIT LOAD EXCEEDS THE WIDTH OF A STANDARD TRAILER AND THE LENGTH IS LESS THAN TRAILER WIDTH THE PROCEDURES AND PRACTICES CONTAINED HEREIN ARE INTENDED FOR VAN TYPE TRAILERS OF ALL LENGTHS THE TRAILERS MAY HAVE WOOD FLOORS METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS

4 THE BLOCKING METHODS SHOWN ON PAGES 3 5 AND 6 ARE FOR WOOD FLOOR TRAILERS WITH UNIT LOADS OF DIMENSIONS THAT PERMIT THE PROCEDURES SHOWN WHEN THE TRAILER HAS METAL FLOORS METAL FLOORS WITH NAILING STRIPS OR THE CHARACTERISTICS OF THE UNIT LOAD MAKES THE BLOCKING AND BRACING SHOWN IMPRACTIBLE TO USE CHOOSE AN APPROPRIATE ALTERNATE METHOD EXPLAINED IN THE SPECIFIC INSTRUCTIONS FOR THIS LOAD

5 A FULL TRUCKLOAD (FTL) CONSISTS OF AS MANY UNIT LOADS THAT CAN BE ARRANGED IN THE TRAILER SINGLE ROW PATTERN CUBE PERMISSIBLE GROSS VEHICLE WEIGHT AND AXLE LOAD LIMITATION PERMITTING IF THESE LIMITATIONS PERMIT UNIT LOADS MAY BE DOUBLE OR TRIPLE LAYER IN ACCORDANCE WITH THE PRINCIPLES SET FORTH IN THIS DOCUMENT

6 A LESS-THAN TRUCKLOAD (LTL) SHOULD BE ARRANGED CONSIDERING WEIGHT DISTRIBUTION IN THE TRAILER A TYPICAL LTL IS SHOWN ON PAGE 6

7 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENT TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA CLOSE AND SEAL TRAILER DOORS AND ATTACH APPROPRIATE PLACARD (IF REQUIRED) TO BOTH SIDES FRONT AND BACK OF TRAILER

8 APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER MM-L-751 NAIL FF-N-100 TYPE II STYLE 10 COMMON BRIGHT STRAPPING QQ S-781 TYPE I CLASS A

9 FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

PROCEDURE (SINGLE LAYER)

THE LOAD SHOWN ON PAGE 3 IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING PROCEDURES FOR A SINGLE LAYER LOAD IN A WOOD FLOOR TRAILER FOR TRAILERS WITH METAL FLOORS OR METAL FLOOR WITH WOOD NAILING STRIPS USE THE APPROPRIATE ALTERNATE BLOCKING METHOD (SEE NOTE 3 BELOW)

WARNING

DO NOT NAIL BLOCKING OR BRACING INTO METAL FLOORS

1 WHEN THE CHARACTERISTICS OF THE UNIT LOAD PREVENTS NAILING SLEEPERS PIECE 4 TO THE TRAILER FLOOR USE A FILLER ASSEMBLY DETAIL L OR M PAGE AS APPROPRIATE ON EACH SIDE OF THE UNIT LOAD

2 WHEN THE TRAILER HAS METAL FLOORS OR METAL FLOORS WITH NAILING STRIPS AND THE STRIPS ARE NOT IN THE DESIRED LOCATION USE A FILLER ASSEMBLY DETAIL L OR M PAGE 7 AS APPROPRIATE ON EACH SIDE OF THE UNIT LOAD

3 THE TYPE OF REAR BLOCKING TO BE USED IS DEPENDENT UPON THE DISTANCE BETWEEN THE TRAILER DOORS WHEN CLOSED AND THE REAR OF THE LADING

DISTANCE LESS THAN 12 INCHES

INSTALL REAR BLOCKING ASSEMBLY DETAIL B AS SHOWN ON PAGE 3

CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED DO NOT USE TRAILERS WITH ROLL UP DOORS

DISTANCE 12 TO 36 INCHES

INSTALL REAR BLOCKING ASSEMBLY DETAIL E PAGE 4 AS SHOWN ON PAGE 5

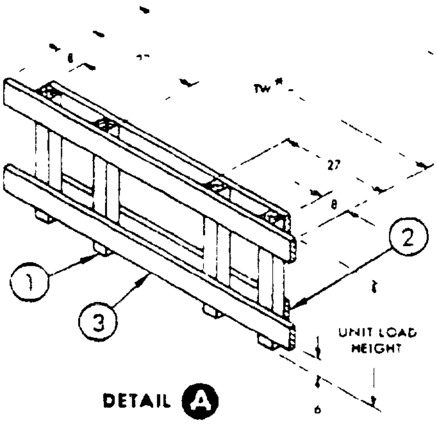
CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED DO NOT USE TRAILERS WITH ROLL UP DOORS

DISTANCE GREATER THAN 36 INCHES

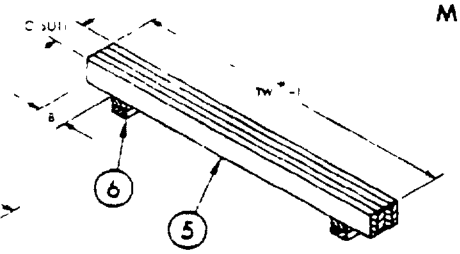
INSTALL REAR BLOCKING AS SHOWN IN TYPICAL LTL PAGE 6 DO NOT USE TRAILERS WITH ALL METAL FLOORS

FILLER ASSEMBLIES SHALL FILL THE VOID BETWEEN THE UNIT LOAD AND TRAILER SIDEWALL


6
27
TW* -
27
8
1
2
3
UNIT LOAD HEIGHT
6

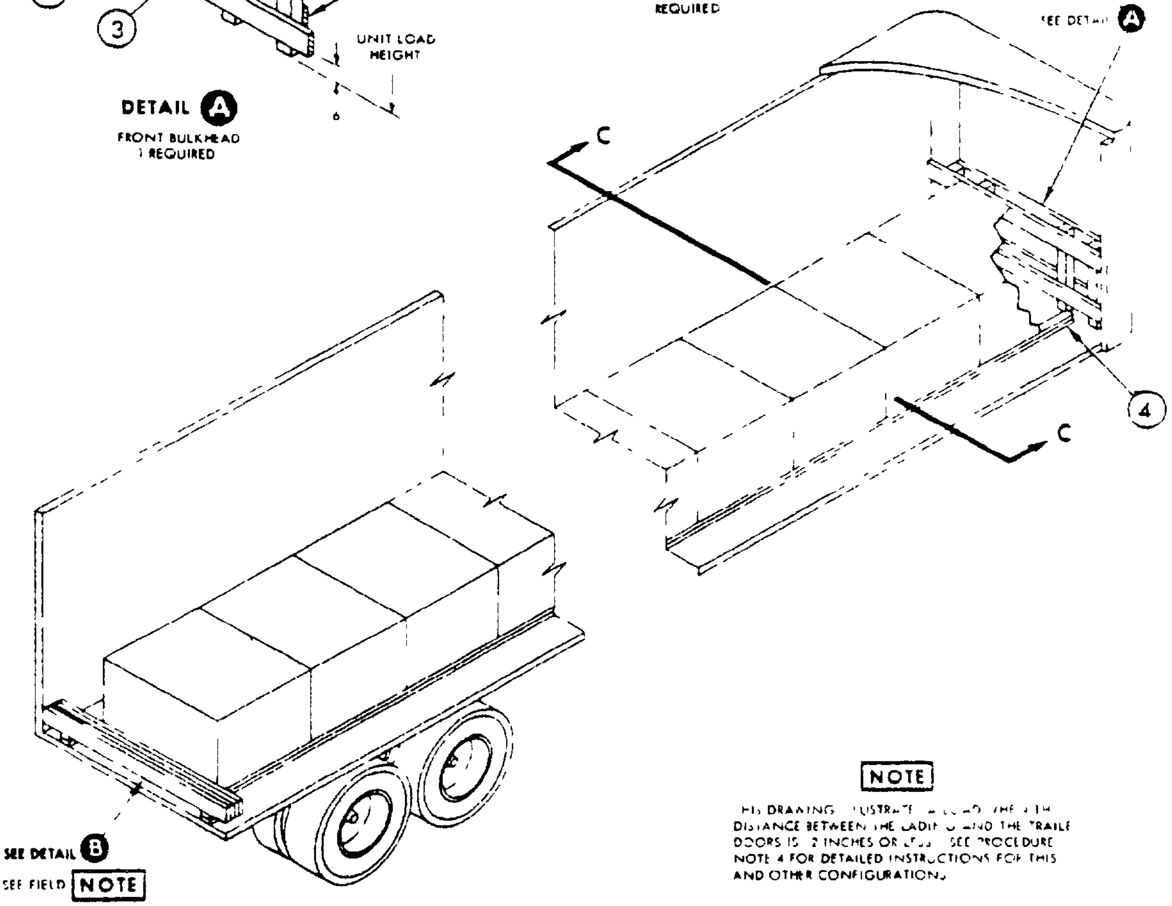
DETAIL A

FRONT BULKHEAD
1 REQUIRED



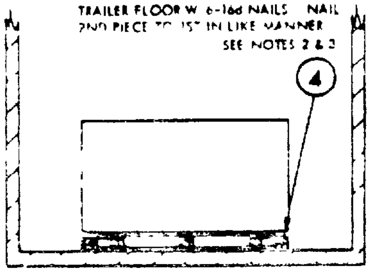
DETAIL B

REAR BLOCKING ASSEMBLY
REQUIRED


SEE DETAIL A
C
C
4
SEE DETAIL B
SEE FIELD NOTE

NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE [illegible] DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS IS 12 INCHES OR LESS. SEE PROCEDURE NOTE 4 FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER CONFIGURATIONS.


2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/ 6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER. SEE NOTES 2 & 3
4

SECTION C-C

*TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL [illegible]	NAIL NUMBER	NAIL SIZE
	[illegible] ORT	[illegible]				
[illegible]	[illegible]	[illegible]	[illegible] REQD	[illegible]	[illegible]	
4	SLEEPER	4 x [illegible]	AS REQD	SEE SECTION C-C		
3	LEFT CROSSMEMBER	2 x 6 x TW*	[illegible]		[illegible] JOINT	2[illegible]
	[illegible] 3	[illegible]				
1	[illegible]	[illegible] UNIT LOAD HEIGHT		[illegible]		

LIST OF MATERIALS AND NAILING DATA

FTL SINGLE LAYER (ALL LENGTH TRAILERS)

PROCEDURE (MULTI-LAYER)

THE LOAD SHOWN ON PAGE 5 IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING PROCEDURES FOR A TWO LAYER LOAD IN A TRAILER WITH ANY TYPE FLOOR (WOOD, METAL OR METAL WITH WOOD NAILING STRIPS).

TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND (AND THIRD) LAYER(S) WHERE THE LAYER CHANGES FROM TWO LAYERS TO ONE LAYER HIGH (OR THREE LAYERS TO TWO LAYERS HIGH) AND AT THE REAR OF THE LOAD WHEN TWO OR MORE LAYERS HIGH, THE UNIT LOAD SHALL BE UNITIZED AS SHOWN ON PAGE 8. THE STIFFENERS SHALL BE POSITIONED TOWARDS THE LOWER LAYER(S) OR, WHEN AT THE REAR OF THE LOAD, TOWARDS THE REAR.

3. FILLER ASSEMBLIES SHALL FILL THE VOID BETWEEN THE LADING AND THE TRAILER SIDE WALLS. WHERE THE LOAD PATTERN IS ONE LAYER HIGH, SLEEPERS MAY BE SUBSTITUTED FOR THE FILLER ASSEMBLY (DETAIL L) PROVIDED THE TRAILER HAS NAILABLE FLOORS AND NAILING CAN BE ACCOMPLISHED. (SEE PIECE 4, PAGE 7.)

4. WHEN SPACE BETWEEN LADING AND TRAILER SIDE WALL IS SMALL, FILLER ASSEMBLIES (DETAILS K AND M) MAY BE SUBSTITUTED FOR FILLER ASSEMBLIES (DETAILS J AND L). (SEE PAGE 7.)

5. *THE TYPE OF REAR BLOCKING TO BE USED IS DEPENDENT UPON THE DISTANCE BETWEEN THE TRAILER DOORS (WHEN CLOSED) AND* THE REAR OF THE LADING.

DISTANCE LESS THAN 12 INCHES

INSTALL REAR BLOCKING ASSEMBLY (DETAIL B) AS SHOWN ON PAGE 3.

CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

DISTANCE 12 TO 36 INCHES

INSTALL REAR BLOCKING ASSEMBLY (DETAIL E PAGE 4) AS SHOWN ON PAGE 5.

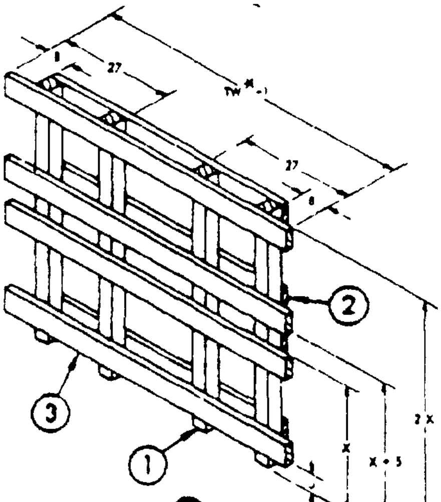
CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

DISTANCE GREATER THAN 36 INCHES

INSTALL REAR BLOCKING AS SHOWN IN TYPICAL LTL, PAGE 6. DO NOT USE TRAILERS WITH ALL METAL FLOORS.

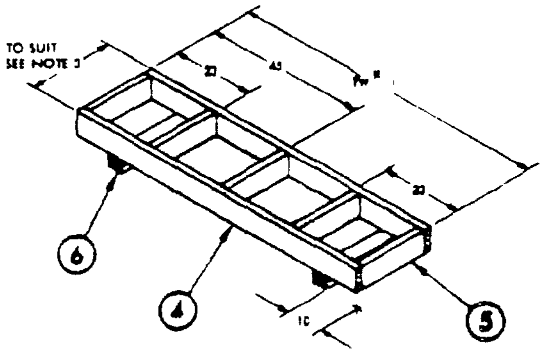
*TW = TRAILER WIDTH



DETAIL D

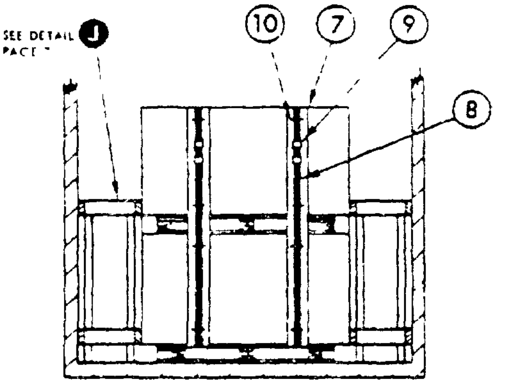
FRONT BULKHEAD

X = UNIT LOAD HEIGHT

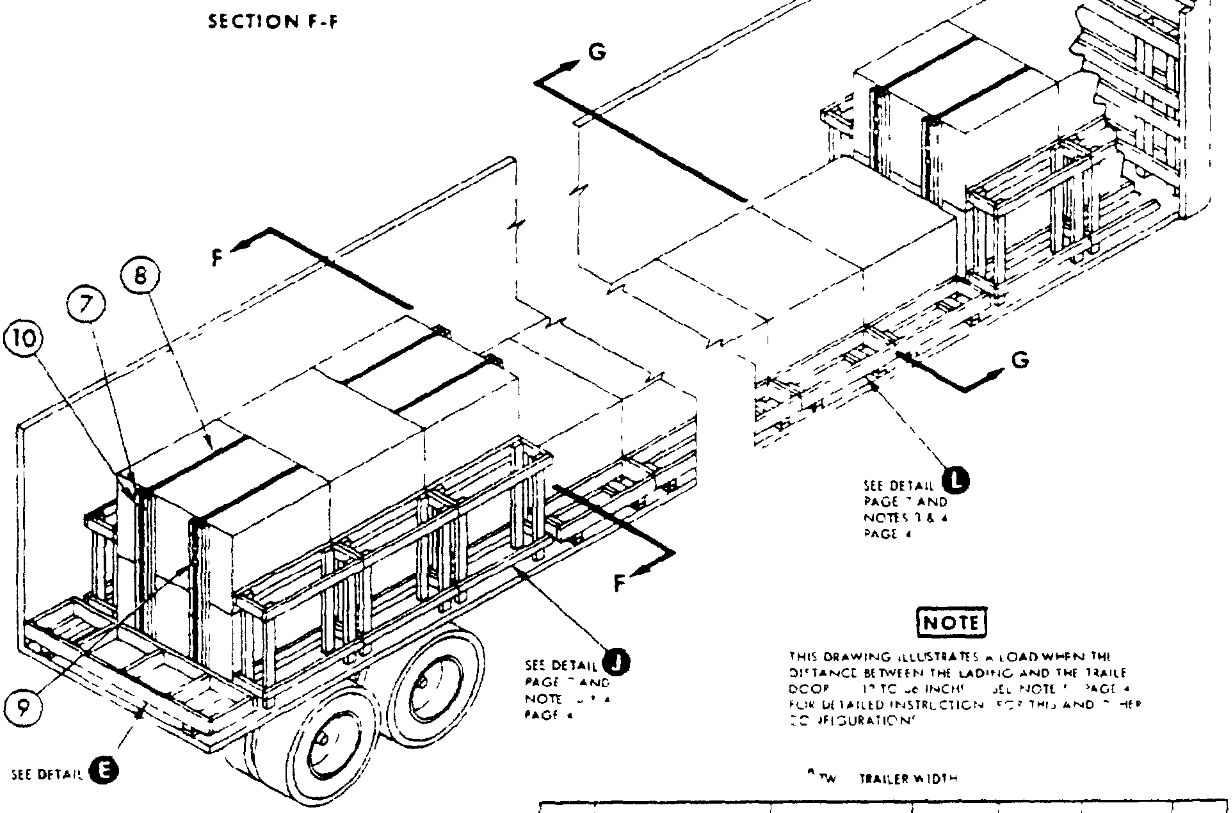


DETAIL E

REAR BLOCKING ASSEMBLY


SEE DETAIL J
PAGE 7
10
7
9
8

SECTION F-F


SEE DETAIL D
G
F
8
7
10
9
G
SEE DETAIL L
PAGE 7 AND
NOTES 3 & 4
PAGE 4
SEE DETAIL J
PAGE 7 AND
NOTE 3 & 4
PAGE 4
F
SEE DETAIL E

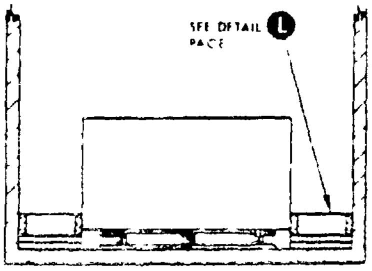
NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOOR [illegible] 12 TO 36 INCHES [illegible] SEE NOTE [illegible] PAGE 4 FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER CONFIGURATIONS

* TW TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
10	STAPLE	FOR 1 1/4 STRAP	24			-
9	SEAL	FOR 1 1/4 STRAP	12			
8	STRAP	[illegible]	6			
7	STIFFENER	[illegible]	12	[illegible] LAMINATE	4	2d
6	SUPPORT	4 TO [illegible]	4			
		[illegible]		[illegible]		
4	CROSSMEMBER	2 x 6 x TW*			JOINT	6d
3	AFT HORIZONTAL	2 x 6 x TW*	4	1	JOINT	[illegible]
2	FWD HORIZONTAL	2 x 6 TO	4		[illegible]	[illegible]
	VERTICAL	4 x 4 [illegible]	4	SEE [illegible]		

LIST OF MATERIALS AND NAILING DATA


SEE DETAIL L
PAGE

SECTION G-G

FTL MULTI-LAYER (ALL LENGTH TRAILERS)

LTL ALL LENGTH TRAILERS

THE LTL SHOWN ON THIS PAGE IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING METHODS FOR LTL IN A WOOD FLOOR OR METAL FLOOR WITH WOOD NAILING STRIPS TRAILERS

WARNING

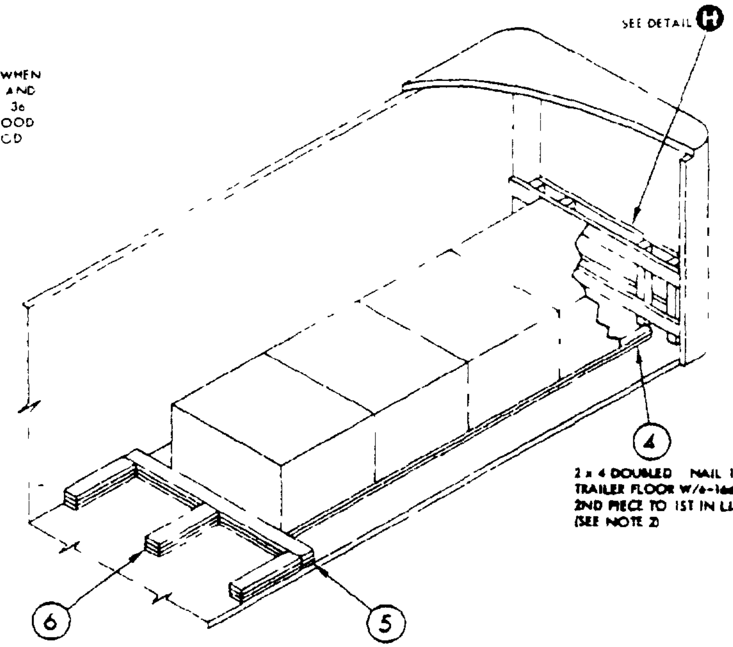
DO NOT USE ALL METAL FLOOR TRAILERS FOR LTL SHIPMENTS AS THEY CAN NOT BE BLOCKED BY THIS METHOD

WHEN THE TRAILER HAS METAL FLOORS WITH WOOD NAILING STRIPS AND THE STRIPS ARE NOT IN THE DESIRED LOCATION USE FILLER ASSEMBLY DETAIL L OR M PAGE 6 AS APPROPRIATE TO FILL THE VOIDS ON EACH SIDE OF THE UNIT LOAD

AN LTL MAY CONSIST OF MORE OR LESS UNIT LOADS AND ARE NOT NECESSARILY LIMITED TO THE AMOUNT SHOWN

NOTE

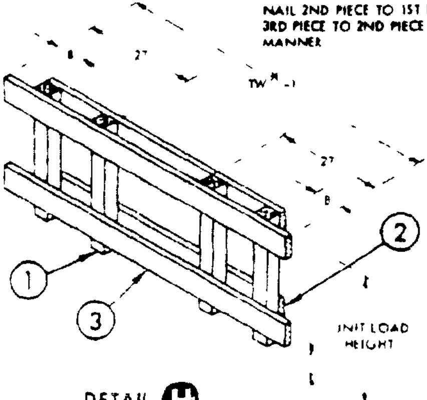
THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS IS GREATER THAN 36 INCHES. THE TRAILER SHALL HAVE WOOD FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS



④ 2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER (SEE NOTE 2)

⑥ 2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER

⑤ 2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/9-16d NAILS. 3RD PIECE TO 2ND PIECE IN LIKE MANNER



DETAIL H

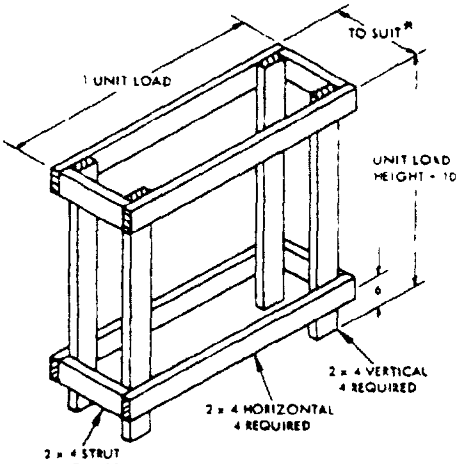
FRONT BULKHEAD REQUIRED

* TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PC REQD	NAILING	NAIL NUMBER	NAIL SIZE
6	BACKUP CLEAT	2 x 6 x 36	[illegible]	SEE FIELD NOTE		
5	CROSSMEMBER	2 x 6 x TW*		SEE FIELD NOTE		
4	SLEEPER	2 x 4 TO SUIT	AS REQD	SEE FIELD NOTE		
3	AFT HORIZONTAL	2 x 6 x TW* - 1			[illegible] JOINT	12d
	[illegible] HORIZONTAL	[illegible]			[illegible]	[illegible]
	VERTICAL	[illegible]	[illegible]			

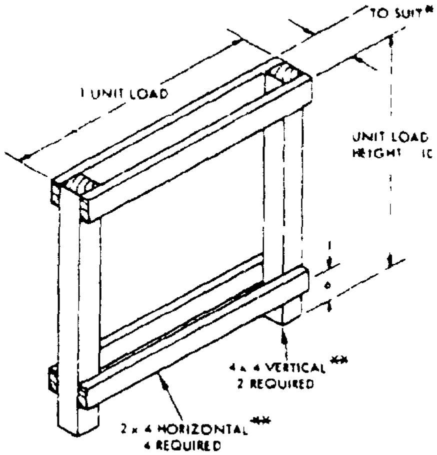
LIST OF MATERIALS AND NAILING DATA

DETAILS (FILLER ASSEMBLIES)


1 UNIT LOAD
TO SUIT*
UNIT LOAD HEIGHT + 10
6
2 x 4 VERTICAL
4 REQUIRED
2 x 4 HORIZONTAL
4 REQUIRED
2 x 4 STRUT
6 REQUIRED

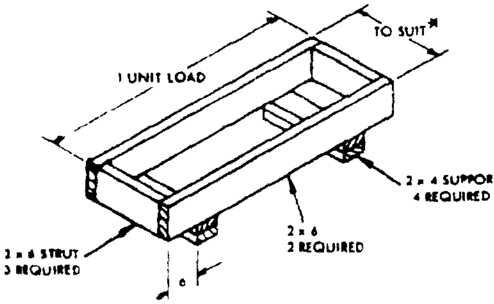
DETAIL J

FILLER ASSEMBLY


1 UNIT LOAD
TO SUIT*
UNIT LOAD HEIGHT 10
6
4 x 4 VERTICAL**
2 REQUIRED
2 x 4 HORIZONTAL**
4 REQUIRED

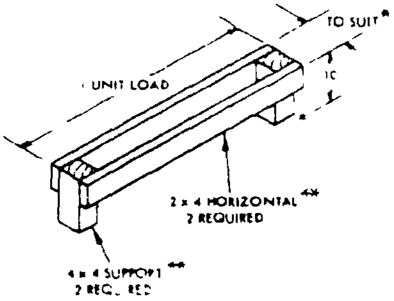
DETAIL K

FILLER ASSEMBLY


1 UNIT LOAD
TO SUIT*
2 x 4 SUPPORT
4 REQUIRED
2 x 6
2 REQUIRED
2 x 6 STRUT
3 REQUIRED
6

DETAIL L

FILLER ASSEMBLY


1 UNIT LOAD
TO SUIT*
10
2 x 4 HORIZONTAL**
2 REQUIRED
4 x 4 SUPPORT**
2 REQUIRED

DETAIL M

FILLER ASSEMBLY

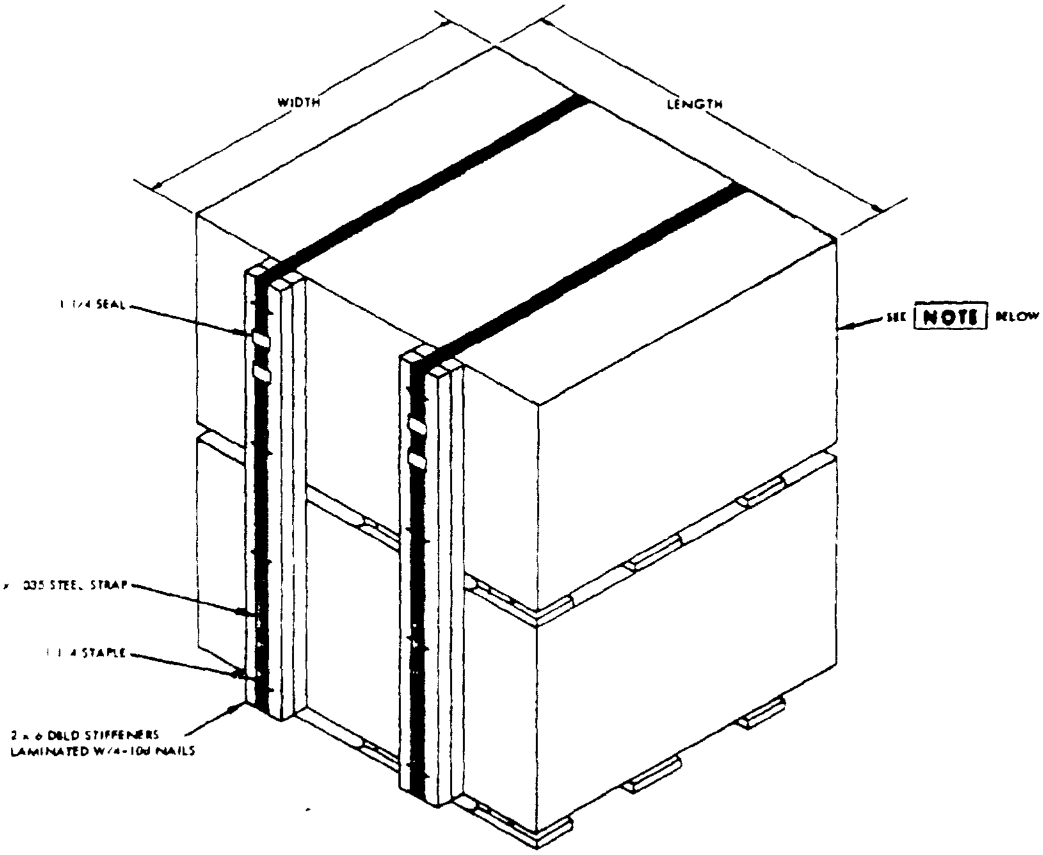
SEE PROCEDURE NOTES 2 & 3 PAGE 2

* SEE PROCEDURE NOTE 4 PAGE 2

** THICKNESS OF [illegible]

MIL-STD-1320-1 (NAVY)

UNITIZING DETAILS


WIDTH
LENGTH
1 1/4 SEAL
SEE NOTE BELOW
1 1/4 x .035 STEEL STRAP
1 1/4 STAPLE
2 x 6 DBLD STIFFENERS
LAMINATED W/4-10d NAILS

1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD, STACKED UNIT LOADS SHALL BE UNITIZED AS SHOWN ABOVE.
2. THE DOUBLED 2 x 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE PALLET OF THE BOTTOM UNIT LOAD.
3. THE 1 1/4 INCH STEEL STRAPS, POSITIONED AS SHOWN, ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE TOP DECK OF THE BOTTOM PALLET.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLE.

NOTE

WHEN LOADING A TRAILER WITH A SINGLE STACK OF UNIT LOADS
THE STIFFENERS SHALL BE AT BOTH SIDES OF THE UNIT LOAD.

5. WHEN UNITIZING THREE HIGH UNIT LOADS, FOLLOW THE PRINCIPLES OF THE TWO HIGH UNIT LOADS SHOWN ABOVE.

REVIEW ACTIVITY
NAVY - [illegible]

PREPARING ACTIVITY
NAVY - [illegible]S
PROJECT NO. 8140-N280[illegible]

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No 22 R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-STD-1320-1 (Navy)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

S/N 0102-014-1801 F-25254

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Colts Neck, New Jersey 07722

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